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## Connection to the WiFlip module

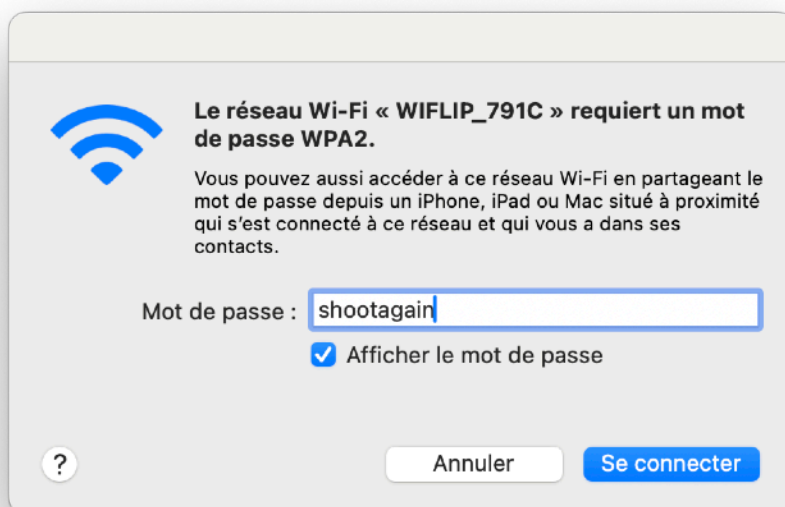
### INITIAL CONFIGURATION

Start by turning on the pinball machine and entering the connection data for your access point (box or router) so that the WiFlip module can connect to it.

To do this, from a computer or smartphone, first find the access point for the WiFlip module. It is called WIFLIP\_XYZT (XYZT varies from a device to another) It is a wifi access point without internet access. If your computer or smartphone cannot see it, you are too far from the WiFlip module. Move closer to it with your computer or smartphone. You can also switch the pinball machine off and on again to restart the WiFlip module.



As your computer or smartphone does not yet know WiFlip, it will ask for a password. This password is : shootagain



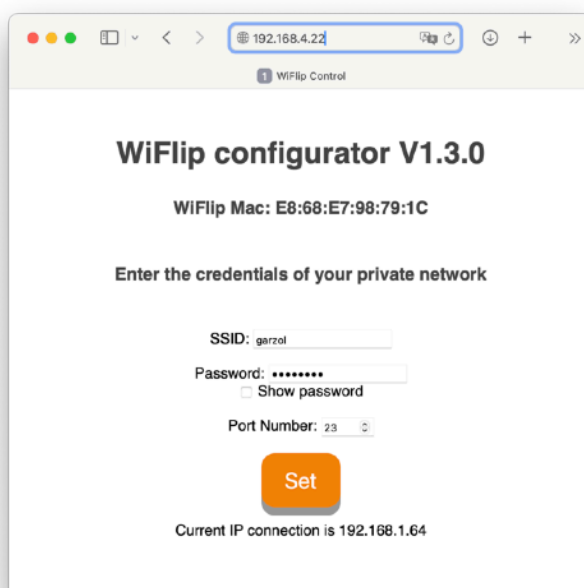
Wait for the connection to be established. Then, in a browser of your choice (Firefox, Safari, etc.), enter the following address in the address bar:

<http://192.168.4.22>

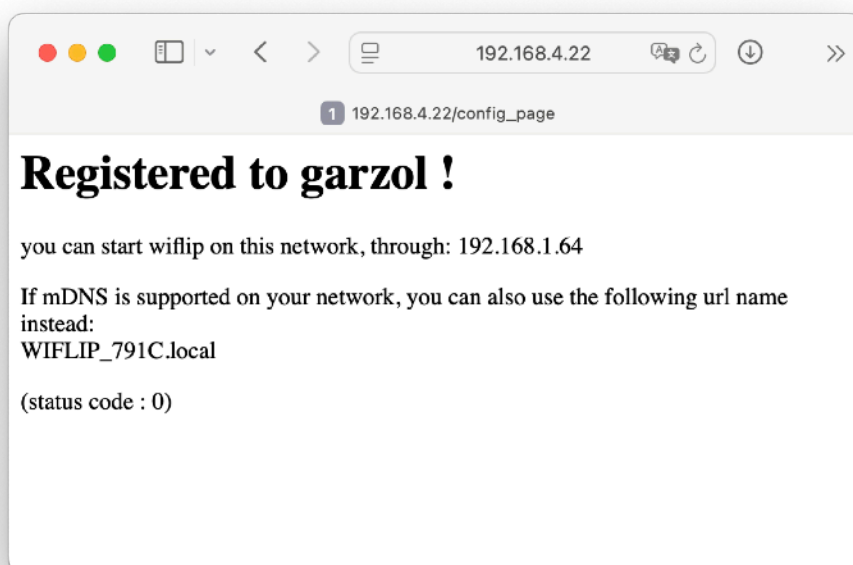
Or [http://wiflip\\_xyzt.local/](http://wiflip_xyzt.local/)

The address starts with http (not https). The connection is therefore not secure.

The following page should appear:



Enter the connection details (SSID/Password) for your box and click on the Set button. After a few moments you should see the following screen:



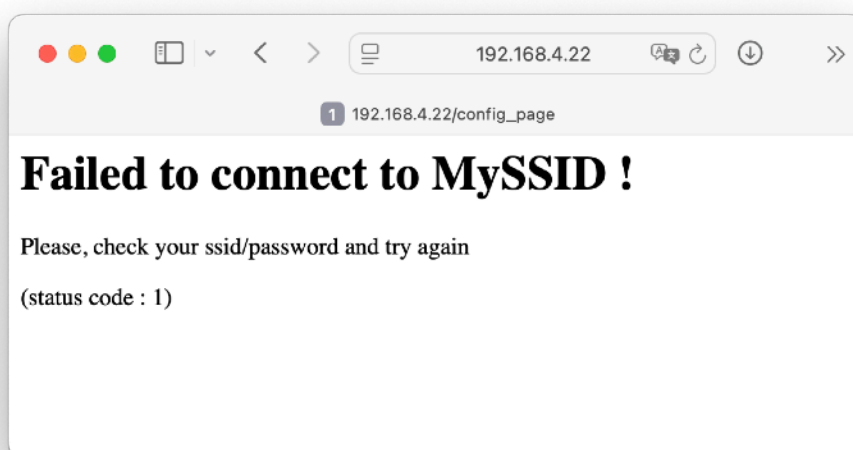
If you have changed the port number you will need to restart the WiFlip module. before the change can be taken into account. Unless you have a good reason for doing so, you should not change this setting.

The login page gives you 2 indications:

1. 192.168.1.64 is the IP address that was given by your DHCP box/router to your WiFlip.
2. WIFLIP\_791C.local is the equivalent name in a more friendly manner. It will never change, while the numeric value is susceptible to change after a box reset for example.

The IP address is susceptible to change. The name in clear will not.

In case of an error, you will see the following page:



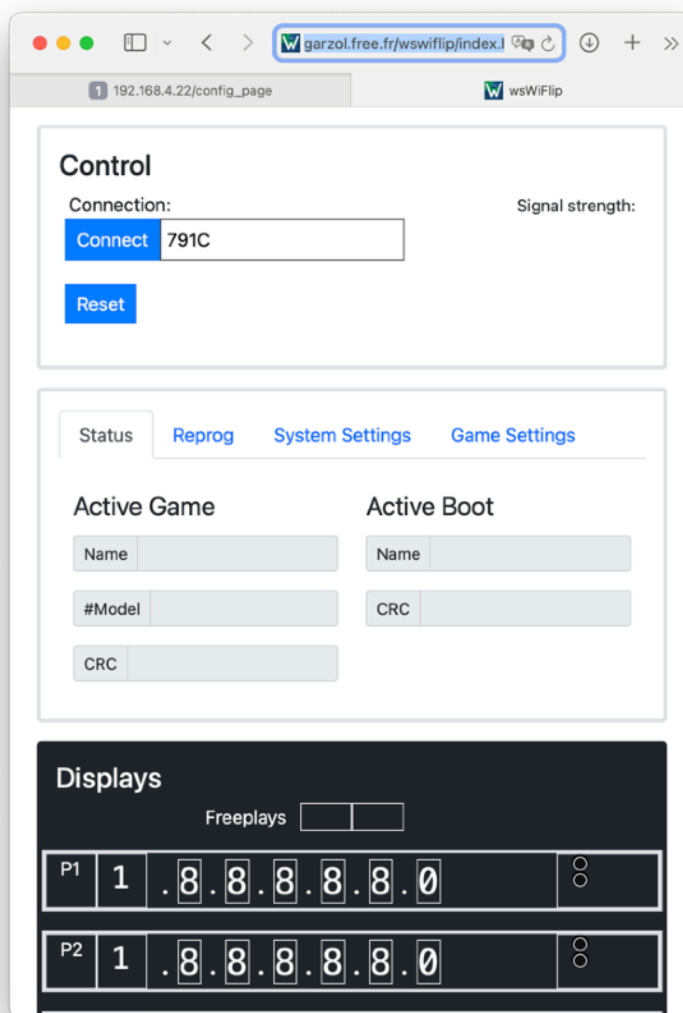
Most of the time, this error page indicates that a wrong SSID or password has been entered. The error may also occur if your box does not support wifi on the 2.4GHz frequency band, which is extremely rare, or, more frequently, if the box, although compatible with 2.4GHz, tries to connect in priority to 5GHz, or 6GHz, by default. WiFlip only supports the 2.4GHz frequency band.

## LAUNCH OF WIFLIP

From a computer or smartphone wifi connected to your box on the same network as the one you chose to configure the WiFlip device, in a browser of your choice, connect to :

<http://garzol.free.fr/wswiflip/>

(Mirrored on <http://aa55.fr/wswiflip/>)



Once again, the connection is not secure. It's up to you to protect your private network with a firewall, for example.

You can open up to 2 simultaneous connections.

You must operate on the same network as the one to which the WiFlip module is connected.

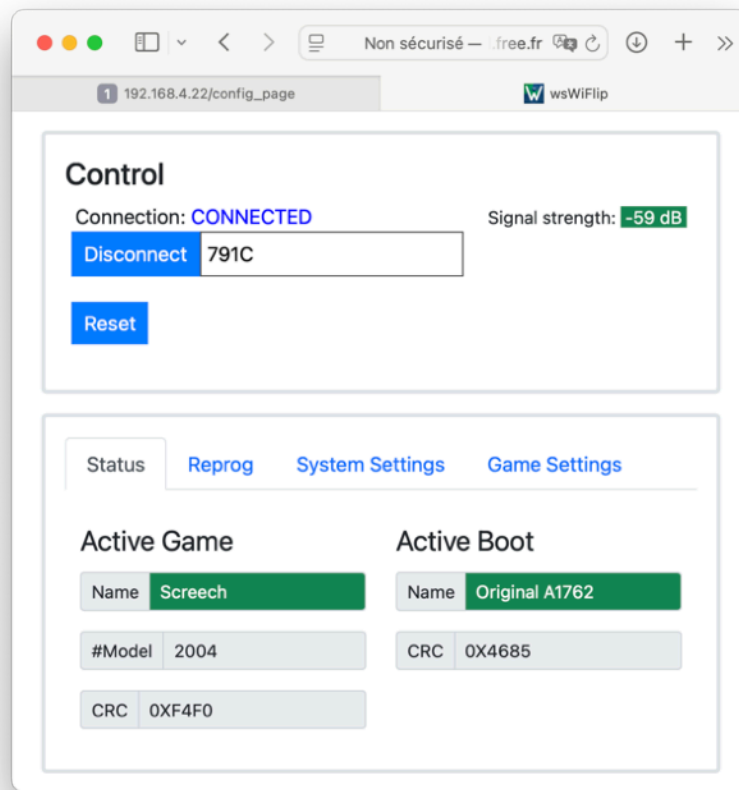
It is perfectly possible to use a wifi hotspot offered by your smartphone.

It is possible to locally save html page (<http://garzol.free.fr/wswiflip/>).

This way, you can run the corresponding script locally. It will work the same way as before. No need for an internet access in this case.

This can be useful if our site is exceptionally down.

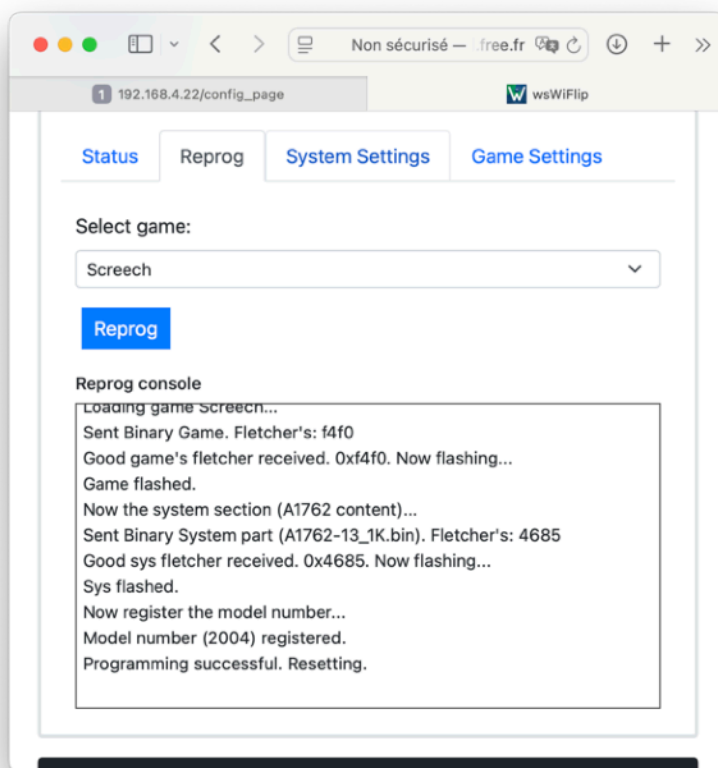
You can now connect to your WiFlip module. Fill in the connection field, either with the raw IP address (such as 192.168.1.64), or with its pretty name (such as WIFLIP\_791C), or just with the last 4 characters (here: 791C). Click the Connect button.



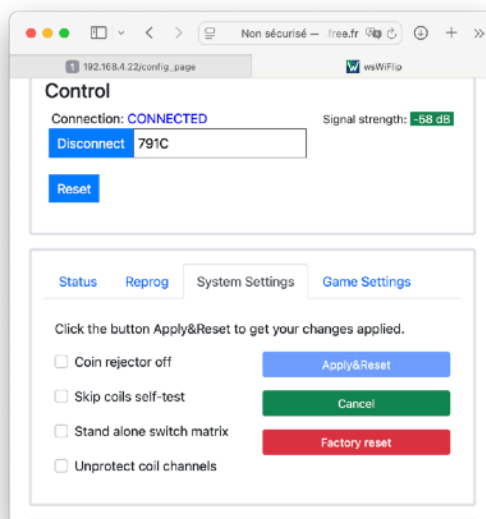
The *Status* tab gives information about the game variant currently running.

The *Reprog* tab lets you choose a new game variant. Simply select the game you want from the list and click Reprog.

After a few seconds, the new game is in place and the MPU reboots. It's ready to go.



The *System Settings* tab lets you configure certain system parameters. For example, you can permanently disable the coin rejector coil. These coils often make an annoying noise, and are not very useful for private use.



**Skip coils self-test:** During startup, the pinball machine checks that the coils are not short-circuited. If this is the case, the sequence is interrupted and an error code appears. The pinball machine locks up. To exit this mode, the problem must be resolved first. For testing purposes, you may want to continue the test sequence without taking this error into account. Check this checkbox in this case.

**Stand alone switch matrix:** Check this box if you are testing the MPU on a bench without connecting it to the MUX board. If the MPU is not connected to the MUX board, all contacts will be seen as “On.” Checking this checkbox reverses the signals that are supposed to come from the switch matrix, and therefore all contacts will be seen as “Off.” If you check this checkbox, you will also need to check *Skip coils self-test* because not only are the switch signals reversed, but also those of the open circuit and short circuit tests, and therefore the initial self-test sequence would be interrupted.

**Unprotect coil channels:** Compared to the original system, an additional layer of coil protection has been added. If activated continuously, a coil may burn out. The added layer cuts off the power to a coil if it remains activated for at least 1 second. The cut lasts 1 second, then the coil is potentially reactivated for a maximum of 1 second and will be cut again, and so on. This saves coils, particularly the ball home ejector coil, which on some game versions remains permanently activated if the ball home switch is stuck in the closed position or broken. To disable this protection, check this checkbox (not recommended).



Factory reset resets the pinball machine to factory conditions. All your settings will be lost. In the Game settings tab, you can change the game settings, such as number of balls per game, high score to date (handicap), etc...

The image shows two screenshots of the wsWiFlip configuration page. The left screenshot displays the 'System Settings' tab, and the right screenshot displays the 'Game Settings' tab. Both pages have a browser address bar showing '192.168.4.22/config\_page' and a 'wsWiFlip' logo.

**System Settings Tab:**

Apply and reset for your changes to take effect.

**Apply&Reset**

|                              |                  |
|------------------------------|------------------|
| 1st Coin Rejector ?          | 2 Plays per Coin |
| 2nd Coin Rejector ?          | 3 Plays per Coin |
| 3rd Coin Rejector ?          | 9                |
| 1st Coin Rejector - #Coins ? | 1 Coin           |
| 2nd Coin Rejector - #Coins ? | 2 Coins          |
| Price Type ?                 | Normal Price     |

**Mode of Play**

|                  |                  |
|------------------|------------------|
| Balls per Play ? | 3 balls per play |
|------------------|------------------|

**Game Settings Tab:**

Apply and reset for your changes to take effect.

**Apply&Reset**

**Score Threshold**

|                |    |
|----------------|----|
| Extra ball ?   | 44 |
| 1st Freeplay ? | 65 |
| 2nd Freeplay ? | 91 |

**Initial Contents**

|               |    |
|---------------|----|
| #Credit ?     | 00 |
| #Replay ?     | 4  |
| #Extra Ball ? | 0  |

Displays replication

The image shows a screenshot of the wsWiFlip 'Displays' configuration page. The browser address bar shows '192.168.4.22/config\_page' and the 'wsWiFlip' logo.

**Displays**

Freeplays 4 0

|    |   |   |   |   |   |   |   |
|----|---|---|---|---|---|---|---|
| P1 | 0 | 6 | 0 | 5 | 1 | 0 | 0 |
| P2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| P3 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| P4 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

Credits 0 7

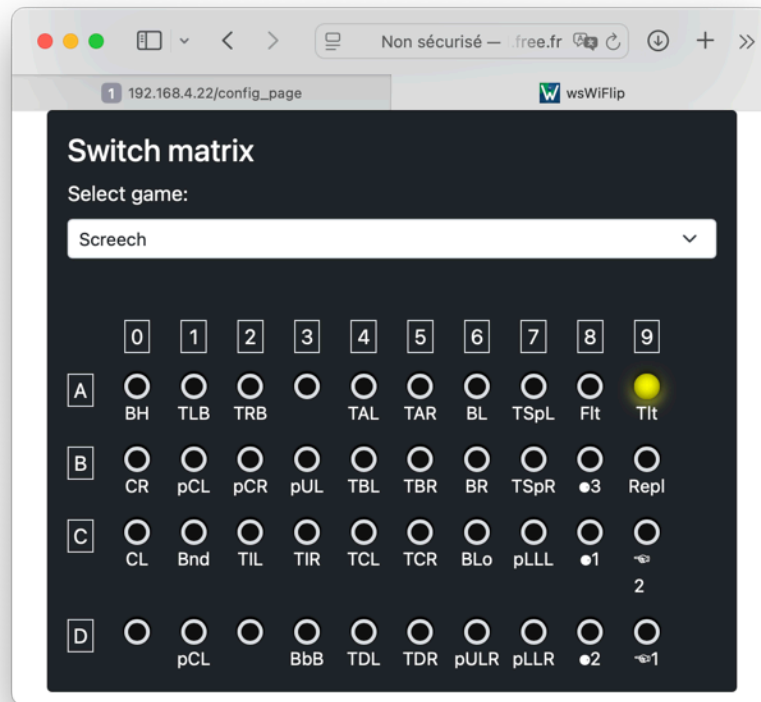
Lottery 7 0

1

### Switch matrix emulation

You can emulate the activation of a switch by clicking on it. The corresponding action is then executed on the pinball machine as if the real switch had been set to the on position.

A double click puts the switch in the on position permanently (in the case of a downed target, for example).



## NETWORK SECURITY CONSIDERATIONS

Network connections come with the risk of hacking. There are two potentially problematic scenarios that you should be aware of:

1. Malicious access to WiFlip anywhere from the internet. With minimal knowledge, from any computer connected to the internet, if your internet box is not properly protected, anyone can easily connect to your pinball machine. If a stranger knows the name of your WiFlip AP (WiFlip\_910A, for example) and the IP address of your router, and nothing is protected, they can launch <http://aa55.fr/wswiflip> from anywhere and connect directly to your pinball machine with entry 910A. Then, anything is possible for them... Change the settings, reset the pinball machine, reprogram it, etc.
2. Direct local access via the `WiFlip_xyzt` access point. This scenario is particularly worrying in cases where your pinball machine is displayed in a public place. Anyone familiar with the WiFlip system can easily scan available networks using their smartphone when they approach a Recel pinball machine. They will then see a network named `WiFlip_xyzt` in the list. As the password is always the same (`shootagain`), it would be easy for them to connect and then, using a browser on their smartphone, go to the page <http://192.168.4.22>. they could easily change the name of the connection point, or even access your pinball machine from the web app <http://aa55.fr/wswiflip/>, activate coils, perform a reset, change settings, and even reprogram a different game version.

### CASE 1 INTERNET ACCESS

It is your responsibility to block incoming connections to the network from outside. Ports 80, 81, and 23, in particular, must be protected, otherwise anyone on the internet will have access to your pinball machine. Typically, you need to make sure that port forwarding on your router blocks these ports. Malicious people still need to have quite a bit of information about your network before they can hack a WiFlip. But you should know that it's not too difficult to do if you don't take the port protection measures mentioned above.

We haven't put in any protection such as certificates or other encryption of the data exchanged to make sure that over time the solution will remain functional (problems with certificate expiry dates, etc...), and to limit unexpected connection problems.

If you want to make absolutely sure that no one will access your machine, just configure the WiFlip device with a non existing WiFi access point. Later, once you need to access it again, reconfigure it with a valid access point. That's it, simple and safe.

### CASE 2 DIRECT ACCESS TO THE WIFLIP AP

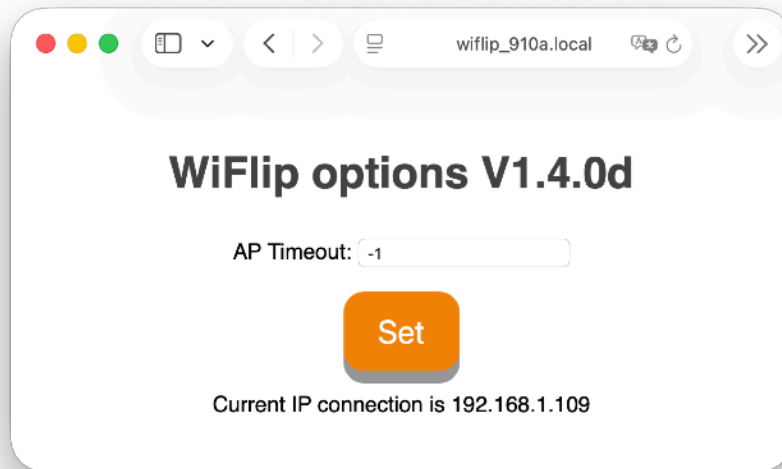
This scenario only makes sense if your pinball machine is on display in a public place.

If your WiFlip version number is 1.4.0 or higher, then you have a solution: Connect to the WiFlip AP and go to the address

<http://192.168.4.22/options>

(or alternatively [http://wiflip\\_xyzt.local/options](http://wiflip_xyzt.local/options))

You should see a page like the one below:



AP Timeout is the delay in milliseconds after which, since the last reset, the AP will shut down and no longer be accessible. -1 means no timeout. If you enter 1000, for example, the AP will shut down after 1 second. This time is short enough that you can no longer connect to it and do anything.

No one will be able to access the WiFlip, including you. That is why it is advisable to use values of around 1 minute to give you time to change this setting again if necessary.

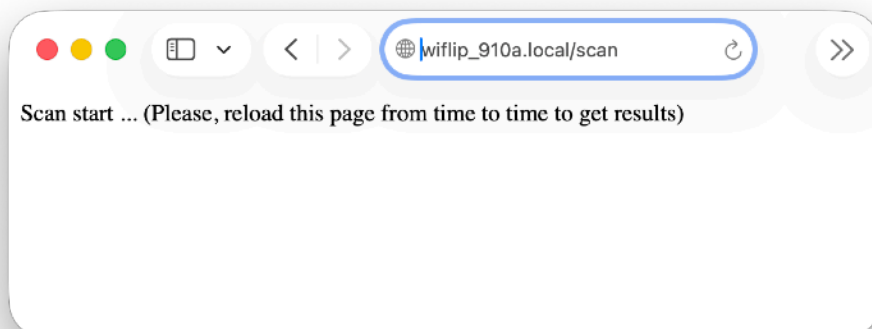
If you accidentally enter a value that is too small, you can always reset the AP Timeout setting to -1 (infinity) by pressing the S2 pushbutton for more than 5 seconds.

## Additional diagnostics.

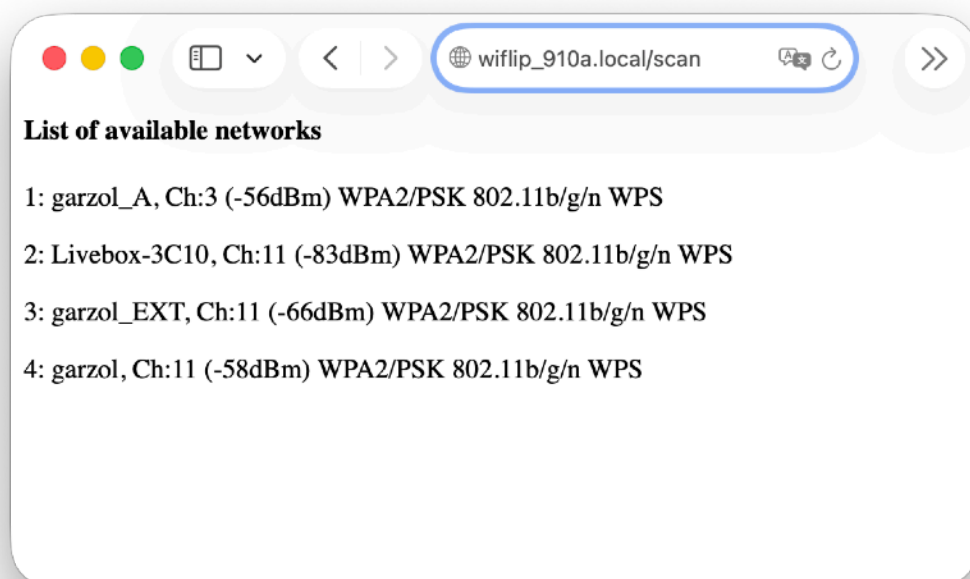
If you have version 1.4.0 or higher, you have access to additional tools.

### NETWORK SCAN

If you are having trouble connecting to your internet box, you can connect to the WiFlip AP and open the page [http://wiflip\\_xyzt.local/scan](http://wiflip_xyzt.local/scan) (or <http://192.168.4.22/scan>).



This will launch a scan of the networks visible to WiFlip. Wait about 30 seconds, then refresh the scan page. You will see a list of available networks:



**LINKS.**

Find out more about WiFlip below:

General infos on the PC or Mac app:

<https://www.pps4.fr/wiflip/>

Online tool address (http only):

<http://garzol.free.fr/wswiflip/index.html>

or

<http://aa55.fr/wswiflip/index.html>

Address of this document:

[https://github.com/garzol/wiflip/blob/main/wswiflip/Doc/userman\\_wiflip\\_en.pdf](https://github.com/garzol/wiflip/blob/main/wswiflip/Doc/userman_wiflip_en.pdf)

### Revision history

| Date       | Version |                                                                                                         |
|------------|---------|---------------------------------------------------------------------------------------------------------|
| 2026-01-25 | V1.3    | Added cybersecurity infos.<br>Added chapter about new features that come with wiflip version from 1.4.0 |
| 2025-10-13 | V1.1    |                                                                                                         |
| 2025-07-10 | V1.0    | Preliminary version                                                                                     |

**CONTACT**

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**NOTES**